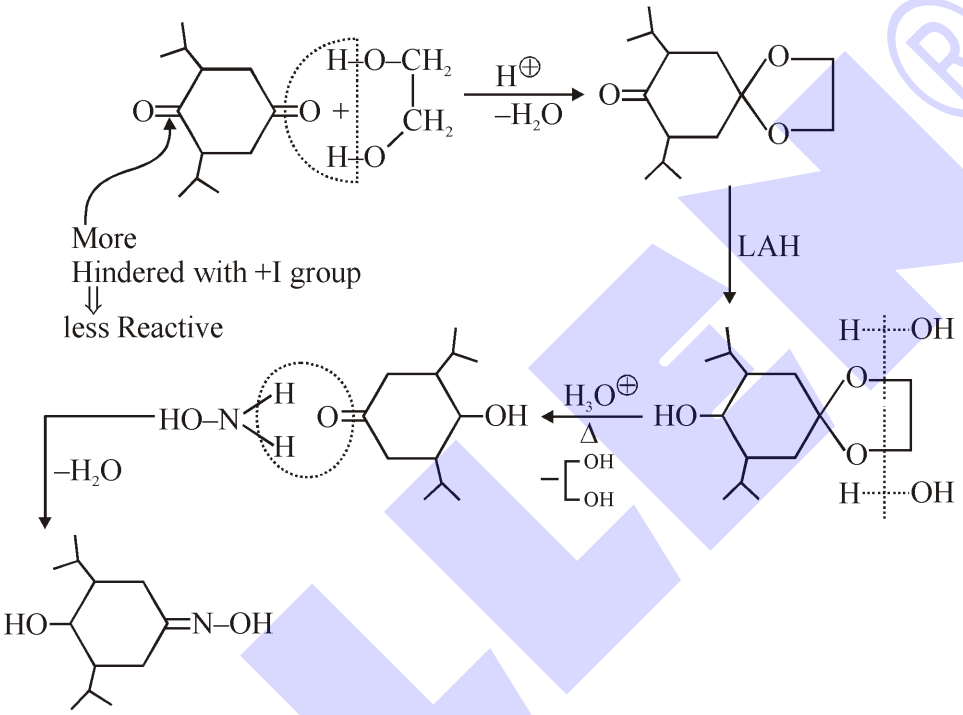
SECTION–I : (i) Only One option correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

3(–1)

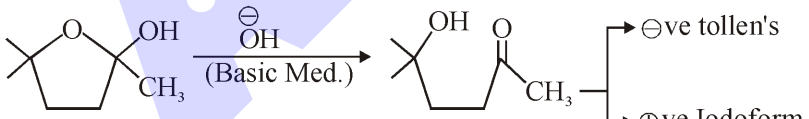
1. (C)

Sol.



2. (B)

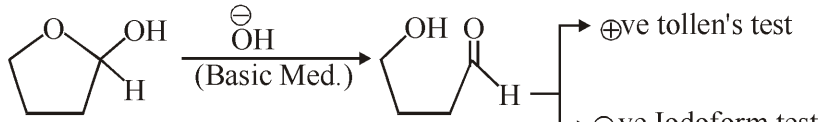
Sol. (I)



[Hemi Ketal]

Comp.-I

(II)



Hemi Acetal

Comp.-II

Note: In both tollen's test and in Iodoform medium is basic and in basic Med. Hemi-ketal & Hemiacetal are opened to corrosponding ketone & Aldelyde

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3. (D)

Sol. (i) Starch – Polymer of α-D-glucose

(ii) glycogen – Polymer of α-D-glucose

(iii) Cellulose – Polymer of β-D-glucose

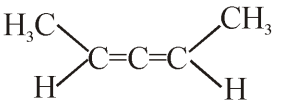
→ On hydrolysis of abvoe polymer for long time they gives their respective monomer unit.

But for prolong treatement both forms of glucose (α & β) are interconvertible into one qnother, hence in mixture, β-D-glucose present in all 3 (A, B & C) mixture

4. (B)

Sol. C_5H_8 → DBE=2 → 1≡ bond is possible or 2= bond is possible

I $CH_2=CH-CH=CH-CH_3$ Show GI ✓

II  Show OI ✓

Relation b/w I & II ⇒ positional Isomer

5. (C)

Sol. $CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH=O + O=CH-CH_2-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-OH$

(1) O_3
(2) Zn/H_2O

$CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH=CH-CH_2-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-OH$ ☒

(i) It is not β–γ undaturated acid

(ii) Mol. formula of x is ⇒ $C_{14}H_{26}O_2$

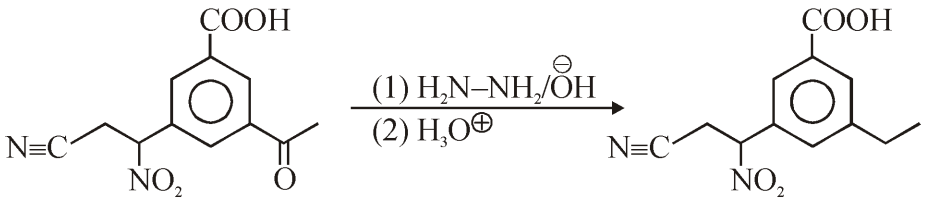
(iii) It shows GI

(iv) It do not shows O.I. hence not form enantiomeric-pair

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6. (C)

Sol.

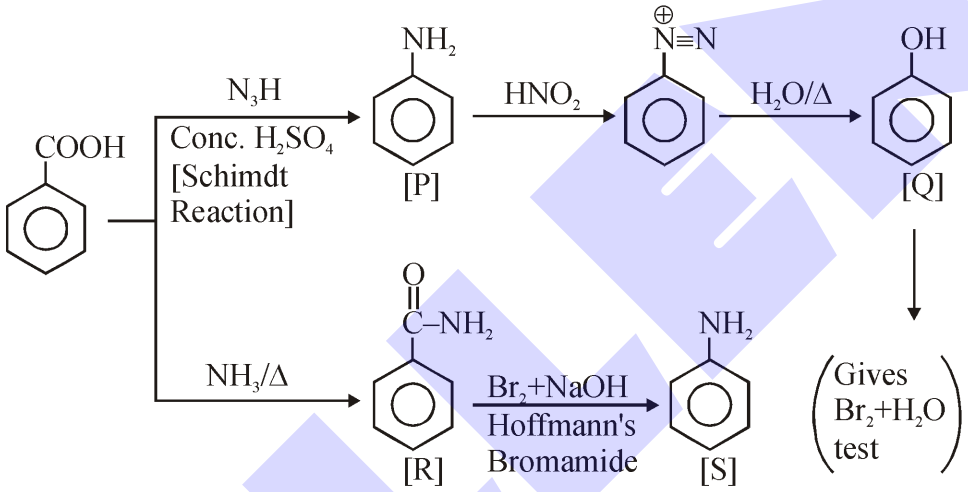


→ In this only ketone group is reduced according to Question thence suitable reagent is wolff kishner reagent because in given example all are acid sensitive so clemmensen reagent is not suitable.

Section-I (ii) [MCQ]

7. (A,B,C,D)

Sol.



All options are correct.

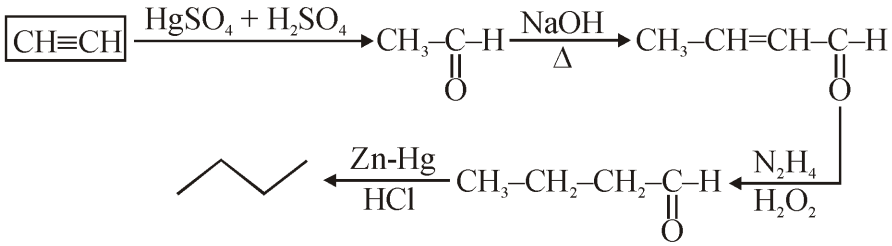
8. (C,D)

Sol. Those Amino Acid which are basic in nature and where more then 1 $-NH_2$ group are present has isolectric point more than 7.

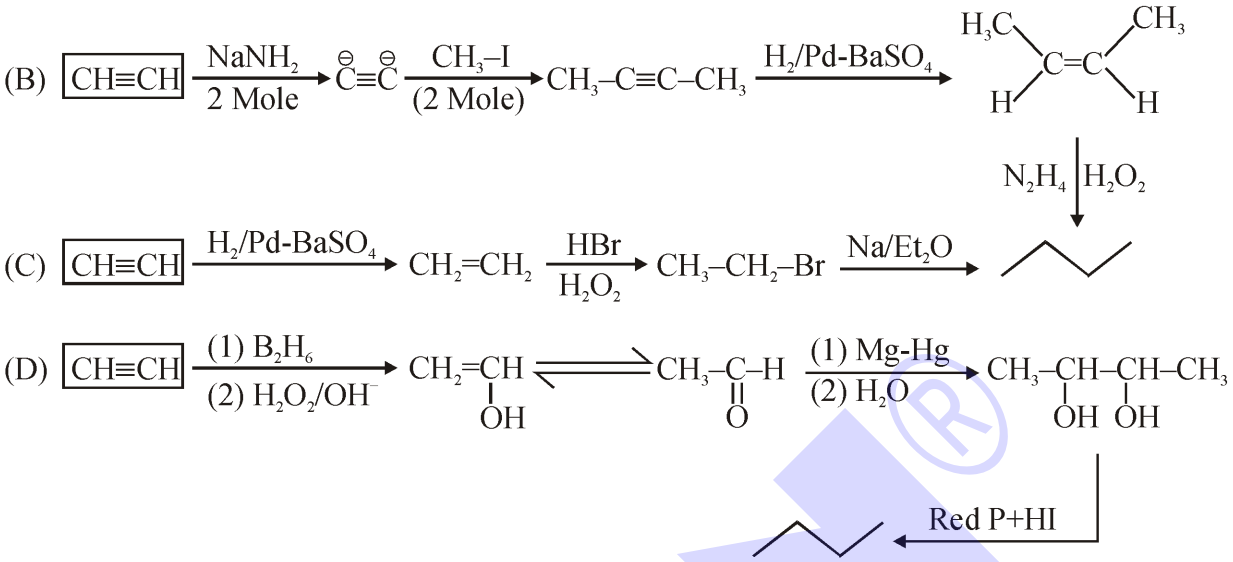
→ Here Arginine and histidine are basic in nature and having more than 7 P^I.

9. (A,B,C,D)

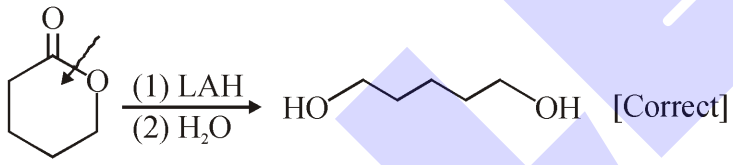
Sol. (A)



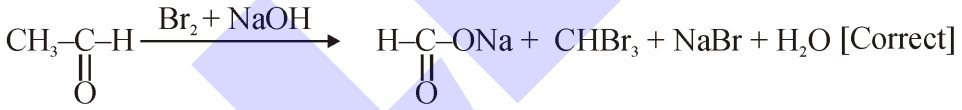
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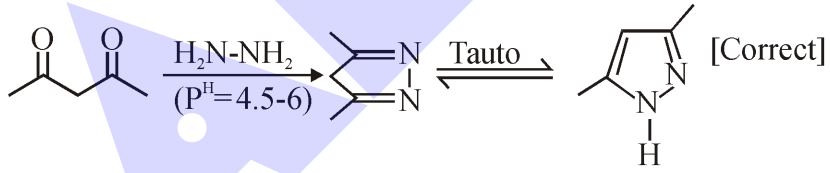
10. (A,B,C)

Sol. **(A)** 

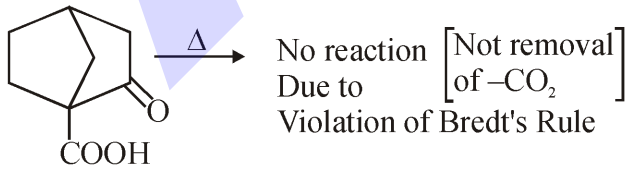
[Correct]

(B) 

[Correct]

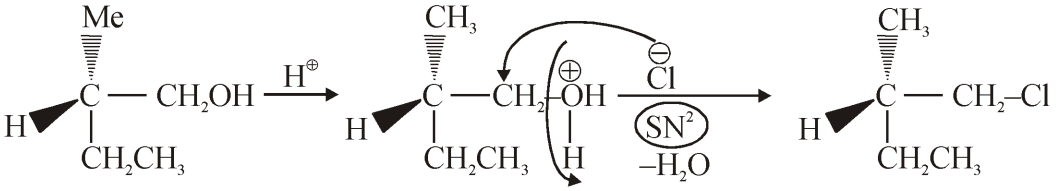
(C) 

[Correct]

(D) 

No reaction Due to Violation of Bredt's Rule [Not removal of $-CO_2$]

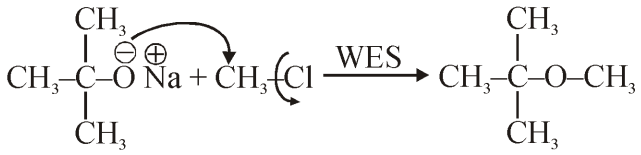
11. (ABCD)

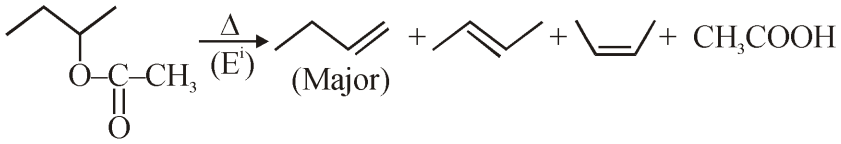
Sol. 

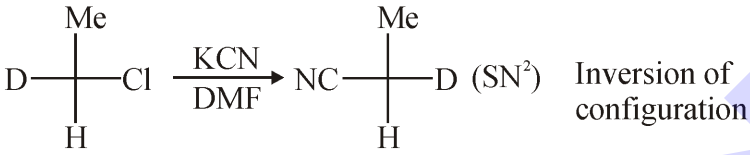
[1° Alcohol]
(1-Chloro-2-methylbutane)

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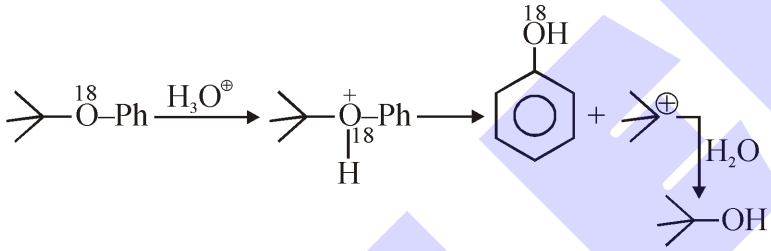
12. (B,C,D)

Sol. (A) 

(B) 

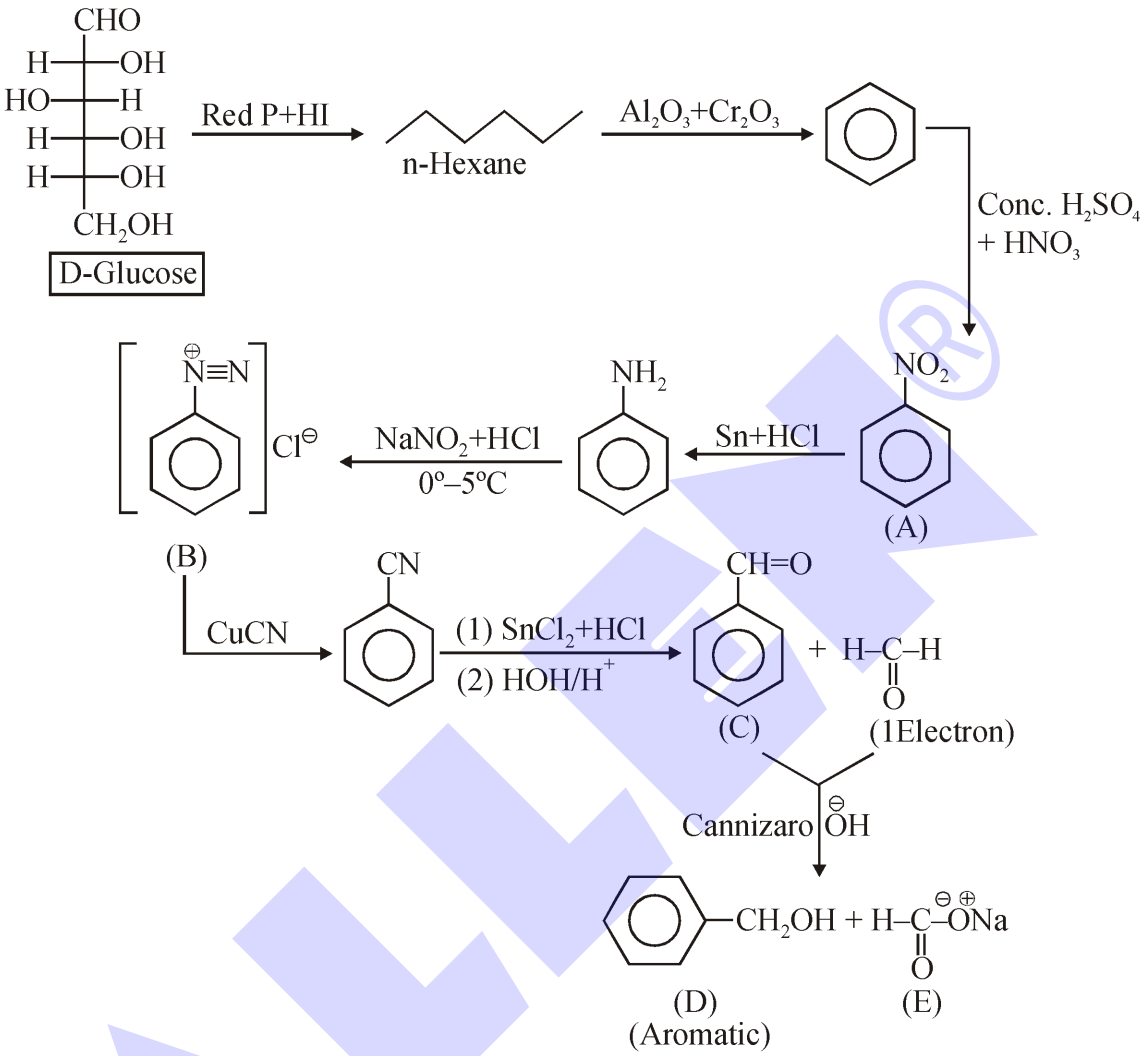
(C) 

Inversion of configuration

(D) 

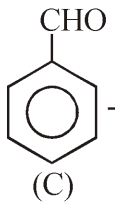
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Paragraph for Q (13) & (14)



13. (A)

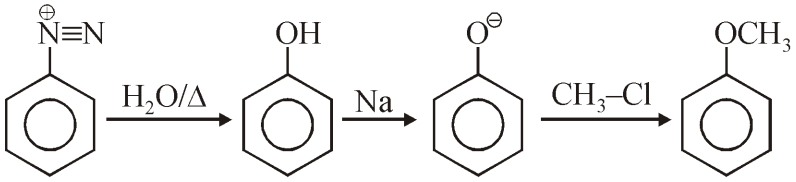
Sol.



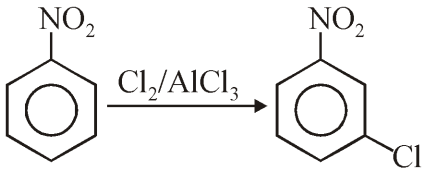
→ Shows cannizaro Reaction and Disproportionation reaction

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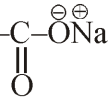
14. (A)

Sol. (A) 

(Correct)

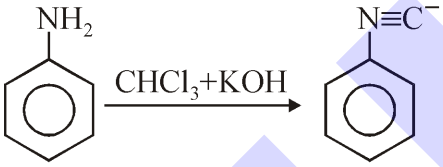
(B) 

(C) Compound [D] is Benzyl alcohol ($PhCH_2OH$)

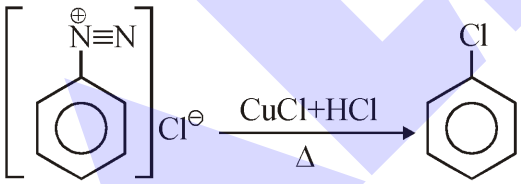
(D) Compound [E] is 

Give B,C,D option are incorrect

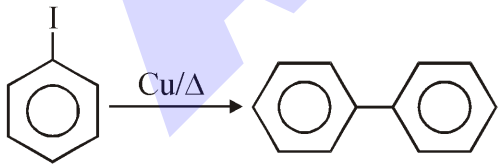
15. (B)

Sol. **(P)** 

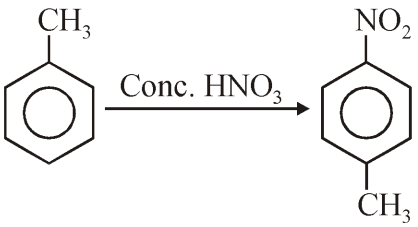
(3)
(Isocyanide test)

(Q) 

(1) (Nucleophillic Substitution in Aromatic)

(R) 

(4)
(Ulmman's Reaction)

(S) 

(2)
(Nitration ESR)

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$\Rightarrow P \rightarrow 3, R \rightarrow 1, R \rightarrow 4, S \rightarrow 2$

Section-III (Integer Type)

1. (7)

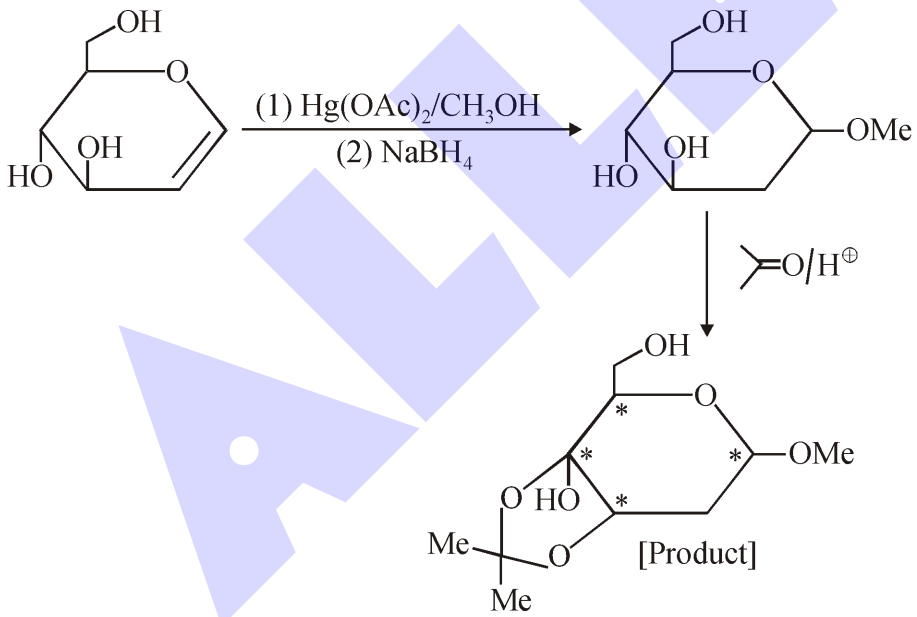
Sol. n-butane $\xrightarrow{Cl_2/h\nu}$ (1) C–C–C–C(Cl)–Cl (2) C–C–C*(Cl)–C(Cl) (3) C–C*(Cl)–C(Cl)–C

(4) C(Cl)–C–C–C(Cl) (5) C–C(Cl)–C(Cl)–C (6) C–C*(Cl)–C*(Cl)–C

Total Dichloro product of n-butane (x) = 10

Value of $x - 3 \Rightarrow 10 - 3 \Rightarrow (7)$

2. **(4)**

Sol. 

Total number of chiral centre in product is (4).

3. **(5)**

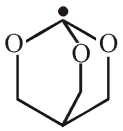
Sol. Concept $[A.S.]_{comp.} > [A.S.]_{H_2CO_3} \Rightarrow$ soluble in $NaHCO_3$ & Release $CO_2\uparrow$

$\rightarrow$ Total (5) compound (a,b,d,g and h) are more acidic then H_2CO_3 hence release $CO_2\uparrow$ gas with

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$NaHCO_3$

4. (3)

Sol. $\dot{C}F_3$ and  are sp^3 hybridised.

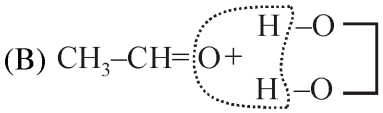
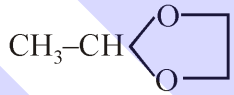
Section -IV [Matric Match]

1. (A) → P,S,T ; (B) → Q ;(C) → R,S,T ;(D) → P,S,T ;

Sol. (A) $C{-}C{=}C{-}C \xrightarrow[(2)\ Zn/H_2O]{(1)\ O_3} CH_3{-}\underset{\underset{O}{\|}}{C}{-}H \xrightarrow{dil.\ NaOH} CH_3{-}CH{=}CH{-}\underset{\underset{O}{\|}}{C}{-}H$

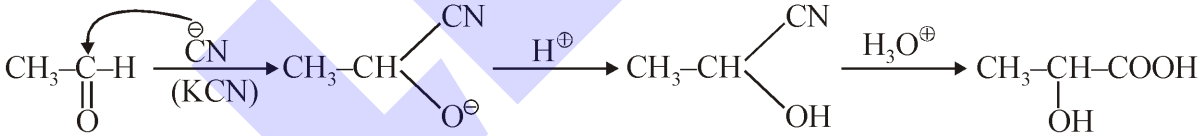
$[(A) \rightarrow P, S, T]$

$\xrightarrow{CrO_3+H_2SO_4}$ $CH_3{-}CH{=}CH{-}\underset{\underset{O}{\|}}{C}{-}OH$

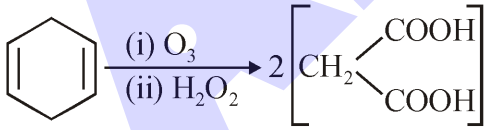
(B)  $\xrightarrow{H^{\oplus}}$ 

(Acetal)→ Cyclic gem diether

$[(B) \rightarrow Q]$

(C) 

$[(C) \rightarrow R, S, T]$

(D) 

$[(D) \rightarrow P, S, T]$

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